

Using Big Data Analytics for Sentiment Analysis to Explore Team Communication Dynamics in Human Machine Interactions for Team Situational Awareness

Vineetha Menon
Department of Computer Science
University of Alabama in
Huntsville
Huntsville, AL, United States
vineetha.menon@uah.edu

Kristin Weger
Department of Psychology
University of Alabama in
Huntsville
Huntsville, AL, United States
kw0004@uah.edu

Bryan Mesmer
Department of ISEEM
University of Alabama in
Huntsville
Huntsville, AL, United States
bryan.mesmer@uah.edu

Sampson Gholston
Department of ISEEM
University of Alabama in
Huntsville
Huntsville, AL, United States
gholsts@uah.edu

Abstract— The team communication dynamics is one of the crucial factors that can provide a plethora of information regarding the evolution of situational awareness and in turn highlight the pivotal moments for mission success in any combat training simulation scenario. Specifically, in this work, we study the team communication dynamics data collected from a fire-fighting simulation. The large volume of communication between individual team members recorded throughout the simulation training provides an ideal basis for the use of Big Data analytics and sentiment analysis to derive deeper insights on various levels of situational awareness observed throughout the simulations. We are interested in understanding the relationship between specific cluster of words or sentiments that are associated with various levels of situational awareness. In particular, we explore the relationship between team communication frequency dynamics, and its relationship to team situational awareness and team performance in a fire-fighting mission success scenario. The outcomes of this research is revolutionary in terms of early identification of team success factors, providing support strategies for improvement of team performance and corresponding team situational awareness in tasks. We experimentally demonstrate that the introduction of Big Data analytics and sentiment analysis in human factor research can invariably boost the situational awareness and team communication dynamics nuances that might otherwise be overlooked. Thus, this research highlights a new avenue for leveraging the team communication dynamics for better situational awareness and mission success objectives.

Keywords—*Artificial Intelligence, Machine Learning, Human Computer interaction, Human-AI interaction, Human Factors, Human Performance Modeling, Human-Machine Interface and Communications, Big Data Analytics*

I. INTRODUCTION

As communication is reflective of team cognition, such as situation awareness, using big data analytics for sentiment analysis can yield meaningful insights regarding crucial team factors that determine mission success. Particularly as high stress environments such as in emergency or combat situations are often challenged by limited communication which hinders coordination efforts and effective decision making [1]. Communication between team members lays the foundation for information sharing and ultimately effective collaboration [2].

At any given point during a mission, team members must understand the current situation and continuously adapt their understanding to the new evolving situation. As such, teams need to be aware of the characteristics of a specific situation such as cues and patterns in the environment [3], as well as team specific characteristics. The degree of situational awareness each team member possesses regarding tasks and performance internal to the team refers to team situation awareness (TSA) [4]. Literature has identified that information exchange concerning TSA impacts coordination efforts and overall mission success [5,6,7,8]. Moreover, precise and effective exchange of information is crucial for a team to update TSA in order to adapt and cope with the ever-evolving situational complexity. As such, teams deployed in complex and dynamic environments must share behaviors regarding their comprehension and projection of internal team situations in order to acquire higher level TSA of internal team characteristics [6,7,8].

Teams may be able to successfully achieve their common goal in complex and dynamic environments if they maintain high levels of TSA among team members through effective team interaction (i.e. team communication and coordination) [4,9]. For instance, during team interaction, team members take turn and shift focus between event phases (i.e., past, present, and future) contingent on the situational context [10]. Communication has been found to be reflective of behaviors related to TSA such as of understanding the teams' situation and action taking [6]. Moreover, sharing of basic factual information about individual team elements (i.e. perception) during the evolving situation was identified more frequently in high-performing teams compared to low performing teams. This was shown through behaviors such as asking and replying to questions related to the team's situation, their current capabilities, and responsibilities. Expanding on communication behaviors that inform about TSA, using big data analysis for sentiment analysis may be appropriate for real-time analysis of team interaction to inform higher-level TSA and decision-making capabilities of the teams.

Given that mission success is driven by the interplay of team communication dynamics, sentiment analysis may be

applied to identify target areas for improvement geared towards higher levels of TSA and team performance in complex and rapidly evolving situations. So far, there have been very few works on sentiment analysis to explore team communication dynamics to identify situation awareness. Sentiment analysis reveals the overall feelings and emotional content inside text data [11]. Primarily, research efforts have focused on the analysis of social media data by extracting users' geo-mapped opinions and sentiments during a disaster to help emergency responders understand the dynamics of the network, e.g., the main users' concerns and panics, the emotional impacts of interactions among users, and the geographical regions that are most affected by the disaster [12]. Moreover, across the majority of the event types studied, Twitter has been the most prevalent platform for situation awareness extraction [13]. For instance, during humanitarian crisis, messages from Twitter platform were filtered and classified in real time to increase the situation awareness. Prior research has also looked at classification systems to identify relevant tweets from disaster related to emergency responses and situational awareness [14]. The classification requirements of situation awareness from tweets are based on the informativeness of its content. This is relevant as more knowledge about the situation at the time of the disaster is directly correlated and can be derived from the disaster related tweets [15]. Further, a study by [16], showed that categorization of tweets by subjectivity, personal or impersonal style, and linguistics (formal or informal) can help predict whether an individual tweet will contain relevant information, with formal and impersonal tweets containing information related to situation awareness.

Ongoing approaches propose advanced methods for classifying sentiment also in microposts, taking typical features of microposts, like re-tweets, hashtags, replies, links, and emoticons, into account [17,18]. Also, several machine learning models have been widely applied to social media in mass emergency, including Support Vector Machine [19,20], Naive Bayes classification [21, 22], random forest [23], LDA [24] and CNN [25,26]. There exists a sound body of literature surrounding AI/ML and big data analytics for sentiment analysis in the context of social media. But, to our best knowledge, there is a significant gap in literature in the use of big data analysis for sentiment analysis in highly dynamic teams where mission success is critical. This paper aims to introduce sentiment analysis for understanding the team dynamics and TSA facets. Hence, this study aims to establish basis for further identification of specific team and individual member communication dynamics as a feedback to improve team performance, AI/ML model learning and human machine interactions specific to the fire-fighting simulation dynamics.

II. PROPOSED METHODS

A. Data Description and Simulation Environment

The dataset consisted of the communication interaction of 42 teams with four members in each team, while carrying out a fire-rescue operation. The participant pool consisted of working professionals and university students who were recruited for this study. The team members did not know each other and were not

familiar with the simulation prior to engaging in their task environment. The fire-rescue simulation environment comprises of fire eruptions exercises and scenarios that would be typically encountered in a real-world fire-rescue operations. This network operation fire-rescue simulation design supports team dynamics studies in the purview of a microworld training. It is also an excellent platform to study human-machine interactions with a focus on teams and team-members to build collaboration and hone in on crisis management skills, team leadership, network-centric command and control operations. The fire-rescue simulation environment challenged team members to co-ordinate for completion of tasks. It also encouraged team member role interdependency and the use of hidden-profile paradigms such as available team member resources, or incomplete information available to individual team members [9]. The fire-rescue simulation design emulated forest and building fires to capture characteristics of a complex adaptive systems (i.e., self-organization and non-linear growth). The fire-rescue operation evolved over time as a function of team members interaction to complete the task. Thus, the fire-rescue simulation used in this study consists of both dynamic and complex environments that captures a multi-level perspective of the team dynamics function.

All team members were assigned independent workstations with a single computer. Each team members was given role and task-specific information prior to the start of the fire-rescue exercise. The teams interacted with one training simulation (simulation 0.0). This was followed by two independent fire-rescue operations: simulation 1.0 and simulation 3.0, respectively, that lasted 15 minutes each. The primary task in each of the simulations was to protect the forest and housing artifacts in the environment from fire eruptions and control the fire eruption situations in the environment. All team members had full visibility of the simulation environment assets such as the terrain, housing, forest and water towers to refill the water tank systems. The simulation 2.0 was a team discussion session that was held in between the fire rescue simulations 1.0 and 3.0, respectively. The simulation data recorded information regarding the individual system capabilities, windspeed and direction, chat window and team performance scores. Each team member was assigned their own fire response system, either a fire-engine or helicopter. Each fire-response system had a different capability such as speed or limited water holding capacity. The variability in task complexity within and across operations was determined by the simulation and the interaction by team members. Throughout the simulation all teams were able to send and receive information in text-based format using the chat window option. At the end of each simulation, teams were presented with an individual performance score, that was based on the percentage of forest and housing saved from fire as a performance feedback. In the study's case, the data was first labeled by two domain expert raters according to the levels of TSA and action taking [6, 27]. The chat logs for all the teams and individual members for the entire duration of the experiment was captured to reflect the large team communication data for sentiment analysis. A collective measure generated from the simulation was used to rate the team performance with respect to completion of the fire-rescue simulation tasks. This team performance rating, and sentiment analysis features extracted were used to train the AI/ML models later in this study.

B. Team Dynamics and Situational Awareness

The derived TSA related information is based on the classification scheme defined by [6]. The derived categories distinguish between perceiving team states or TSA factors (perception), comprehending relevant facts perceived within the team (comprehension), projecting relevant future team states or factors (projection), and exchanging information regarding team member's action as described in Table I.

TABLE I. TEAM DYNAMICS, TSA CATEGORIES ALONG WITH THEIR DEFINITION AND EXAMPLES OF DATA COLLECTED (ADAPTED FROM [6])

Code #	Team Situation Awareness	Definition	Examples
3	Perception	Information about team factors and their current state such as condition, modes, action.	"My water tank is empty"
4	Comprehension	Information about task related occurrences within the team that help other team members understand team relations, team events and places.	"Our fire engines seem to be slower than our helicopters"
5	Projection	Information about possible future actions in regards to own team.	"We will have to split up areas"
6	Action	Action of team members in regards to own team.	"Fire engines fill up your water tank"

Fig. 1. gives an example of team communication data collected from the fire-rescue simulation, where team member numbers denote the four individual team members (1-4), team member communications via chat was recorded along with their timestamps for the entire experiment. Thus, constituting a vast amount of sentiment data to be analyzed in 42 teams. Four TSA code levels from Table I that were exhibited by teams were recorded during the course of the experiment, where TSA code 3 indicates 'perception', code 4 stands for 'comprehension', code 5 represents 'projection' and code 6 demonstrates 'action'. Each of these TSA codes were hand annotated based on the coding scheme presented in Table I. If discrepancies in the coding occurred, both domain expert raters came to a complete agreement. There was substantial agreement between coders $\alpha = .776$ (95% CI, .734 - .810, and $p < .001$).

Members	Total	Simulation	SA Team
3	0:01:39	Don't be stealing my fire.	0 6,3
2	0:01:49	fire on the left	0
2	0:02:50	break :)	0
4	0:03:00	No fire	0
1	0:03:17	the suspense is killing me	0
2	0:03:24	fire	0
3	0:03:40	where?	0

Fig. 1. A pictorial representation of the team communication dynamics simulation data

C. Sentiment Analysis

In this work, two categories of features were extracted from the dataset excerpt shown in Fig. 1 as follows:

- Observed team dynamics features: These features were directly extracted from the experimental data to include the:
 - Number of turns (communication messages) by the team
 - Number of turns taken (communication messages) by each team member
 - The TSA rankings (code 3-6) given to each turn
 - The TSA rankings (code 3-6) for each member in the team

- Derived team dynamics features: These team features were calculated using the observed features in aggregate for each simulation such as:
 - Mean TSA of the team
 - Mean TSA of the team members
 - Frequency of each TSA ranking

Sentiment analysis on the body of team communication data was performed using the valence aware dictionary for sentiment reasoning (VADER) model in the NLTK package. The large volumes of team communication messages were analyzed using the VADER sentiment intensity analyzer. Each message was assigned a sentiment intensity score based on their associated emotions for positivity, negativity, neutrality, and a compound score were extracted through combination of all the three scores to capture complex nuances in sentiments that map to varied TSA levels. For instance, the sentence in Table I for a TSA code of 3 is a neutral sentiment or establishes facts, whereas TSA code of 6 sentence reflects a more active sentiment about the situation. The mean values for each sentiment intensity score across each simulation were used as team dynamics lexicon features along with the team performance simulation rating based on TSA to train the AI/ML models for prediction of team performance. Fig. 2 illustrates the schematic representation of the proposed sentiment analysis-based AI/ML architecture to predict the team performance.

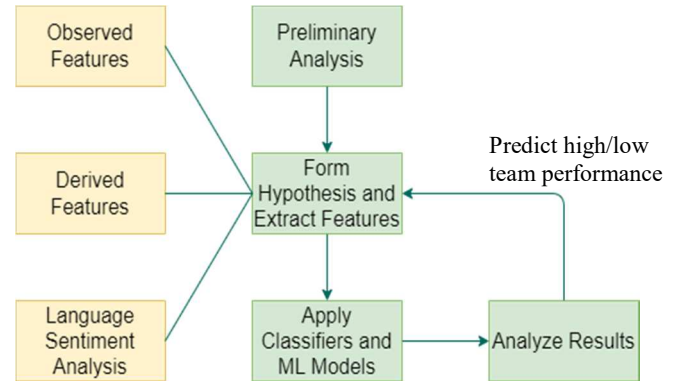


Fig. 2. Proposed sentiment analysis based AI/ML architecture to predict the team performance based on the team dynamics

D. AI/ML Supervised Classification Models

As depicted in Fig.2. the collective team dynamics features, i.e., the observed, and derived team dynamics features along with the language sentiment scores features were used to train four AI/ML techniques that gave the best team performance predictions for our study is discussed as follows:

1) *Decision Tree*: Decision Trees are one of the intuitive tree-based structured and commonly supervised learning algorithms for classification and regression ML applications [23, 28]. The decision trees predict the value of a target variable through simple if-then-else decision rules that are learnt from the data features. Deeper decision trees imply more complex decision rules and better AI/ML model generalization capabilities. The root node of the decision tree denotes the entire team dynamics data, while the leaf nodes denote the team performance classification or prediction decision. The navigation path of nodes/branches in the decision tree represent the specific team dynamics feature choices made to arrive at a particular team dynamics classification or prediction decision. Thus, the various paths in the decision tree can intuitively convey insights into which team dynamic features play a significant role in understanding and capturing the team performance dynamics.

2) *Random Forest*: It is another widely used tree based supervised ML technique for classification and regression tasks. It is an ensemble learning extension of decision trees, where multiple optimized decision trees are used to build an ensemble ML model [23, 28]. The final prediction performance of the random forest technique is dictated by the team performance class labels (high, medium and low classes) that are commonly selected by the greatest number of decision trees.

3) *Naive Bayes*: Naive Bayes is one of the simplest and widely used traditional ML supervised algorithm for classification and prediction applications. It is based on the famous Bayes theory, and follows the principle of maximum aposteriori probability (MAP) estimation for a probabilistic data classification and prediction [21, 22]. The posterior class probability of a test data instance is calculated using class conditional density estimation and class prior probability parameters and is assigned to the class with the maximum posterior class probability.

4) *Support Vector Machines (SVM)*: SVM is one of the state-of-the-art ML algorithms that are extensively used in various classification and prediction applications [19,20]. SVMs are a statistical based supervised learning technique that aims to find an optimal hyperplane to distinctly separate the data into different team performance classes. In this work, we employ a linear kernel based SVM classification model, due to its best classification performance.

5) *Multilayer perceptron (MLP)*: The MLP is an artificial neural network that belongs to the family of feedforward learning algorithms [25,26,29]. MLPs typically constitute three different categories of layers—the input layer, output layer and the hidden layer. The input layer processes the input team dynamics data as a linear weighted combination that is subject to an RELU activation function. The final prediction and classification tasks are performed by the output layer. In each hidden layer, the weighted linear combination data learning is propagated (feed forward) from one hidden layer to another. These in between hidden layers in the MLP are responsible for heavy computation, data abstraction, feature extraction and

complex data learning rocesses. In this work, we employed an MLP network with 2 hidden layers, 100 neurons per layer, with learning rate of 0.001 and the ADAM optimizer was used.

III. EXPERIMENTAL RESULTS AND DISCUSSION

The preliminary analysis consisted of identification of relevant observed and derived team dynamics features that was used to predict the team performance from the experimental data. A histogram analysis was conducted to study the relationship between the TSA and team performance. The TSA scores were analyzed using the sum of the median and mode TSA scores for each simulation. All 42 teams in the experimental data, were evenly split based on their TSA scores into three performance groups (14 teams each) of high, mid and low performance teams. The top 33% (14 teams) with high TSA scores were ranked as high performance teams, the next 33% (14 teams) with moderate TSA scores were ranked as mid performance, and the last 33% teams with low TSA scores were ranked as low performance. TSA threshold for each team performance tier was empirically chosen. Fig. 3. illustrates an instance of the histogram analysis conducted on all teams to generate the frequency of various TSA scores for each category of simulations (0-4) to understand the influence of TSA on team performance. Note that a cumulative TSA score of 10 is observed in the low performance teams. This cumulative score is based on the TSA code of 6 (action), and 4 (comprehension), that we simultaneously exhibited in the team dynamics. Likewise, a cumulative TSA score of 9 in team 5, denotes the cumulative of TSA code 6 (action), and 3 (perception).

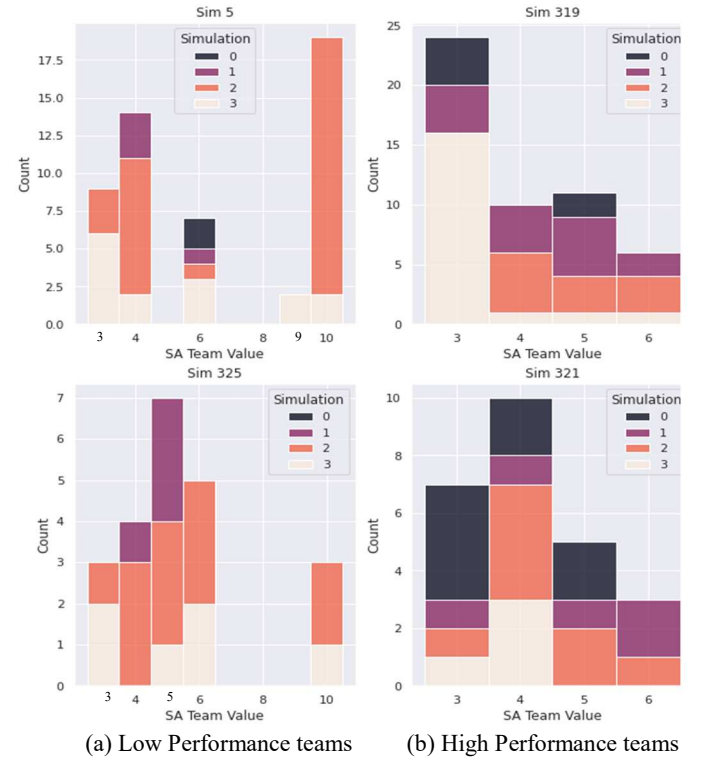
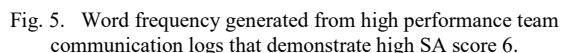
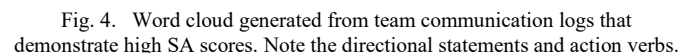


Fig. 3. Comparison of the TSA ranking over various simulation tasks. Low performance teams (left). High performance teams (right).

In addition, we performed sentiment analysis on the team communication data logs of all teams and generated word clouds to visualize the most frequently occurring words during the course of simulation as depicted in fig.4. The motivation behind word cloud analysis is to identify if there are a specific category of words that can be associated with better team dynamics and TSA coding levels. From fig. 4, it can be observed that nouns such as ‘fire’, ‘truck’, ‘forest’, ‘area’ etc. were used more often considering the experiment was indeed a task to extinguish fires. We also note a mixture of noun and verbs in the context of action and directional team dynamics words such as ‘take’, ‘think’, ‘top’, ‘left’, ‘need’, ‘help’, etc. are identified as the second category of most occurring words.



communication dynamics such as ‘take’, ‘right’, ‘left’, ‘help’ etc. Whereas, similar words occur in lower usage frequency in the low performance team communication logs. Thus, it can be implied that active team member interactions while demonstrating a high level of TSA throughout the different simulation tasks is a good indicator of the high performance teams dynamics. Hence, from figs. 3-6, we can postulate that appropriate proactive simulation training strategies or exercises could benefit enhanced human-machine interaction experience and higher TSA levels for improved team performance.

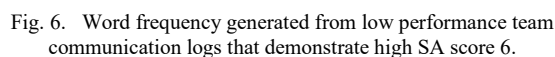


TABLE II. CLASSIFICATION ACCURACIES ACROSS MULTIPLE TRAINING RATIOS USING SENTIMENT ANALYSIS FEATURES.

Methods		% Training Data		
	70%	60%	50%	40%
Decision Tree	75.86	81.50	65.00	50.00
Random Forest	66.67	75.00	85.00	62.50
Naïve Bayes	75.12	81.25	75.00	66.67
SVM	75.45	68.75	65.00	50.00
MLP	75.86	68.77	75.00	45.83

Therefore, this study confirms that understanding team dynamics and TSA indeed requires a multifaceted big data analysis approach. It proves the potential of using big data analytics for sentiment analysis to gain deeper insights on active team communication, TSA and overall team dynamics factors

that contribute to better human machine interactions and team performance in fire-fighting simulation scenario. The aim of this study is to open new collaborative AI/ML, big data analytics and human machine interaction research avenues in SA, team performance dynamics and human factor areas.

IV. CONCLUSION

In this case study, we present a multi-faceted big data analytics-based sentiment analysis framework for team performance prediction. We conducted a complex analysis of the relationship between various individual, team communication dynamics and TSA factors to understand their role in influencing the overall team performance success in the task of putting out fires. The complex big data analysis based on the team communication logs, team performance dynamics and their situational awareness aspects can provide meaningful insights on how to improve team interactions in a wide variety of simulation and real-world scenarios. Especially, sentiment analysis has great potential to provide targeted areas for improvement in TSA, team interaction and performance dynamics as a feedback mechanism to improve AI/ML model training as well as for better human machine interaction dynamics in the fire-fighting simulation. A limitation of this case study is that supervised learning methods need labeled data to learn the model and are strongly influenced by the use of TSA features (feature engineering). Future work will delve deeper into effects of automated generation of TSA and team performance labels for sentiment analysis. We will also explore the influence of various individual and team performance factors for improved TSA in various simulated scenarios.

REFERENCES

- [1] E. Martin, I. Nolte, E. Vitolo, "The four Cs of disaster partnering: Communication, cooperation, coordination and collaboration," *Disasters*, 2006; 40(4): 621-643
- [2] S. L. Marlow, C.N. Lacerenza, J. Paoletti, C. S. Burke, and E. Salas, "Does team communication represent a one-size-fits-all approach?: A meta-analysis of team communication and performance," *Organizational Behavior and Human Decision Processes* 144: 145–170, 2018.
- [3] N. J. Cooke, E. Salas, J. A. Cannon-Bowers, and R. J. Stout, "Measuring team knowledge," *Human factors*, 42(1), 151-173, 2000.
- [4] M.R. Endsley, W.M. Jones, "Situation awareness information dominance and information warfare," Technical Report No. AL/CF-TR-1997-0156. Wright- Patterson Air Force Base, OH: U.S. Air Force, Armstrong Laboratory. 1997.
- [5] K. Weger, "A classification of behavior indicators exhibited in team communication: The intricate role of team reflexivity with team cognition, regulatory processes and performance," University of Bamberg Press. 2018
- [6] K. Weger, V. Menon, A. Hollingsworth, and H. Schaub, "Information Sharing Behaviors in Text-based Communication during Emergency Response", *Journal of Emergency Management*, 2022, in press.
- [7] M.R. Endsley, "Toward a theory of situation awareness in dynamic systems," *Human Factors*. 1995; 37: 32–64.
- [8] M. Demir, N.J. McNeese, N.J. Cooke, "Team situation awareness within the context of human autonomy teaming," *Cognitive Systems Research*. 2017; 46(3): 12.
- [9] J. C. Gorman, N.J. Cooke, and J.L. Winner, "Measuring team situation awareness in decentralized command and control systems," *Ergonomics*, 49, 1312–1325, 2006.
- [10] K. Weger, S. Leder, B. Mesmer, V. Menon, and H. Schaub, "How Effectively Do We Communicate? An Analysis of Team Reflexivity in Transition and Action Phases during Team Collaboration," *IEEE Transactions on Professional Communication*, 2022, 65(3). doi 10.1109/TPC.2022.3186773
- [11] A. Karami, V. Shah, R. Vaezi, and A. Bansal, "Twitter speaks: A case of national disaster situational awareness," *Journal of Information Science*, 46(3), 313-324, 2020.
- [12] V. K. Neppalli, C. Caragea, A. Squicciarini, A. Tapia, and S. Stehle, "Sentiment analysis during Hurricane Sandy in emergency response," *International journal of disaster risk reduction*, 21, 213-222, 2017
- [13] J. Vongkusolkrit, and Q. Huang, "Situational awareness extraction: a comprehensive review of social media data classification during natural hazards," *Annals of GIS*, 27(1), 5-28, 2021
- [14] W. Maharani, "Sentiment analysis during Jakarta flood for emergency responses and situational awareness in disaster management using BERT," In proceedings of 8th IEEE International Conference on Information and Communication Technology (ICoICT), 2020, pp. 1-5.
- [15] L. J. Moreira and L. A. Silva, "Prototype generation using selforganizing maps for informativeness-based classifier," *Computational intelligence and neuroscience*, vol. 2017, 2017.
- [16] S. Verma, S. Vieweg, W. Corvey, L. Palen, J. Martin, M. Palmer, and K. Anderson, "Natural language processing to the rescue? extracting" situational awareness" tweets during mass emergency," In Proceedings of the International AAAI Conference on Web and Social Media, Vol. 5, No. 1, pp. 385-392, 2011.
- [17] Barbosa, L., & Feng, J. (2010, August). Robust sentiment detection on twitter from biased and noisy data. In *Coling 2010: Posters* (pp. 36-44).
- [18] A. Agarwal, B. Xie, I. Vovsha, O. Rambow, and R. J. Passonneau, "Sentiment analysis of twitter data," In Proceedings of the workshop on language in social media (LSM 2011), pp. 30-38, 2011.
- [19] M. Imran, C. Castillo, F. Diaz, and S. Vieweg, "Processing social media messages in mass emergency: A survey," *ACM Computing Surveys (CSUR)*, vol. 47, no. 4, pp. 1–38, 2015.
- [20] Imran, M., Mitra, P., & Castillo, C. (2016). "Twitter as a lifeline: Human-annotated twitter corpora for NLP of crisis-related messages". *arXiv preprint arXiv:1605.05894*.
- [21] A. Goel, J. Gautam, and S. Kumar, "Real time sentiment analysis of tweets using Naive Bayes," In 2nd IEEE International Conference on Next Generation Computing Technologies (NGCT), pp. 257-261, 2016
- [22] H. Parveen, and S. Pandey, "Sentiment analysis on Twitter Data-set using Naive Bayes algorithm" In 2016 IEEE 2nd international conference on applied and theoretical computing and communication technology (iCATccT) pp. 416-419, 2016.
- [23] T.K. Ho, "Random Decision Forests". Proceedings of the 3rd International Conference on Document Analysis and Recognition, Montreal, QC, 14–16 August 1995. pp. 278–282.
- [24] B. Resch, F. Uslander, and C. Havas, "Combining machine-learning topic models and spatiotemporal analysis of social media data for disaster footprint and damage assessment," *Cartography and Geographic Information Science*, vol. 45, no. 4, pp. 362–376, 2018.
- [25] F. Alam, S. Joty, and M. Imran, "Graph based semi-supervised learning with convolution neural networks to classify crisis related tweets," in Twelfth International AAAI Conference on Web and Social Media, 2018.
- [26] F. Alam, F. Ofli, M. Imran, and M. Aupetit, "A twitter tale of three hurricanes: Harvey, Irma, and Maria," *arXiv preprint arXiv:1805.05144*, 2018.
- [27] Taboada M, Brooke J, Tofiloski M et al. Lexicon-based methods for sentiment analysis. *Computational linguistics* 2011; 37(2): 267–307.
- [28] T. Dietterich, "An Experimental Comparison of Three Methods for Constructing Ensembles of Decision Trees: Bagging, Boosting, and Randomization". *Machine Learning*. 40 (2): 139–157, 2000.
- [29] Y. LeCun, Y. Bengio, and G. Hinton, Deep learning. *Nature* 521, 436–444 (2015)